



Position also matters! Separating Same Instruments in String Quartet using Timbral and Positional Cues

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Abstract

String quartets inherently face challenges such as fundamental frequency overlap, similar timbres, and complex acoustic environments, making existing source separation methods relying solely on timbral characteristics less effective. This paper proposes a novel approach that enhances separation performance by encoding timbral and spatial positional cues to guide a neural network for source separation. To evaluate our method, we created a dataset of string quartet recordings with realistic reverberation with two different seating arrangements. Experiments demonstrate that our approach outperforms current state-of-the-art (SOTA) methods in source separation, achieving superior performance even in seating arrangements different from training, thus highlighting the importance of our timbral and positional encoding in improving separation quality and generalizability.

Index Terms: music source separation, classical music, string quartet, conditional source separation

1. Introduction

Music source separation (MSS) has made significant progress with the advent of deep learning, enabling the extraction of individual sound sources from complex audio mixtures [1, 2, 3, 4, 5, 6]. However, the current development in this field has been heavily influenced by the limitations of existing datasets, such as MUSDB18-HQ [7], MedleyDB [8], and FMA [9], which primarily feature modern pop and rock music styles. As a result, most music source separation research is focused on the Sony De-mixing Challenge [10], where the goal is to separate individual stems—such as vocals, bass, drums, and other instruments—after they have been mixed and subjected to various non-linear effects like gain, delay, and reverb. In practice, it has been proven feasible using advanced deep learning architectures, such as BS-Roformer [6], Demucs [3], and Band-split RNN [11]. However, these methods typically rely on separating stems based on distinct timbres, thus losing their effectiveness when it comes to separating instruments with similar timbres.

In chamber music, due to the demands of composition, there exists multiple instances of homogeneous musical instruments (e.g., more than one violin) and instruments from the same family (e.g., the violin family, including violins, violas, and cellos). These instruments share similar physical structures, leading to close timbres and significant spectral overlap, which greatly increases the difficulty of source separation [12]. Moreover, unlike the linear summation inverse operations used in pop or rock music separation, the natural reverberation in concert halls introduces a nonlinear mixing process [13, 14, 15]. This nonlinearity further complicates the source separation task.

Separating sources in chamber music settings offers several

promising practical applications. It can enhance the quality and efficiency of live recordings by improving post-production capabilities, allowing selective re-recording of specific parts, and reducing overall recording costs. Additionally, it holds the potential for transforming traditional stereo recordings into immersive 3D surround sound experiences by accurately extracting individual sound sources [16]. From a social welfare perspective, it could also benefit individuals with hearing loss by enabling personalized adjustments of instrument volumes, as demonstrated in projects like the Cadenza initiative [17]. These practical applications highlight the need for more advanced separation methods tailored to the unique acoustic challenges of chamber music, such as those presented by string quartets.

Motivated by above, in this paper, we focus on string quartet (consisting of two violins, a viola and a cello), the typical situation of chamber music, to explore source separation with similar timbre instruments under complex reverberant conditions. As shown in Figure 1, although the arrangement of these instruments is uncertain, the position candidates are fixed, which provides valuable positional cues that can be leveraged to improve separation performance. Specifically, we incorporate timbral and positional cues to guide a neural network for source separation, achieving results of each instrument in the string quartet, which as far as we know, has not been investigated yet. Our main contributions are as follows:

- We introduce a new dataset: String Quartet with Reverberation Set (SQR Set) by collecting data for string quartets, comprising data synthesized using a Sound Library (for training and validation) and individual instrument audio recordings (for testing), and simulated real performance seating scenes using real-world sampled B-format hall reverb with two kinds of seating arrangements.
- We develop a source separation system that can separate homogeneous musical instruments by incorporating timbre and positional cues into a modulation network, demonstrating its effectiveness in handling complex acoustic environments.
- We train our proposed method only on one seating arrangement and test it on another. The experimental results demonstrate that the separation performance of our method remains almost unaffected. In contrast, other state-of-the-art (SOTA) methods used for comparison showed almost no applicability in this scenario. This validates the effectiveness and generalizability of our method.

The remainder of this paper is organized as follows: Section 2 reviews related work in music source separation. Section 3 presents our proposed method, detailing timbral and positional information dynamic embedding. Section 4 describe the experimental setup and evaluate the performance of our method. Section 5 concludes the paper.

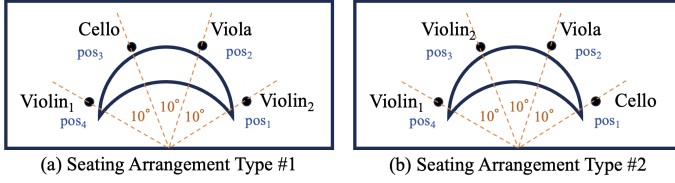


Figure 1: *String Quartet Seating Arrangement Types*

2. Related Works

2.1. Music Source Separation

Music source separation aims to extract different sound sources from a single audio mixture. Current methods for music source separation can be broadly categorized into three approaches based on how the model relates to the sources:

- **Individual Model for Each Source:** This approach trains a separate model for each source, resulting in multiple models corresponding to the number of sources. While this method can be highly effective for specific sources, it is computationally expensive and not scalable for many sources. Examples include BS-RoFormer [18] and Band-Split RNN [4].
- **Shared Model for All Sources:** A single model separates all sources simultaneously, outputting multiple tracks in one pass. This method is more scalable and efficient but may struggle with distinguishing between sources when their characteristics are highly similar. Demucs [1] and its subsequent improvements [19, 3] exemplify this approach.
- **Shared Model with Conditional Control for Each Source:** In this approach, a single model is employed for all sources, but it outputs different source tracks by applying specific conditions. This method allows for flexibility and improved separation quality by leveraging prior knowledge or context about the sources. Examples include MSI [20] and LaSAFT [21].

2.2. Chamber Ensemble Source Separation

Research on source separation for chamber music is relatively limited. Lin et al. propose a conditional source separation method using the timbre of the target instrument as a query based on the URMP dataset [20, 22]. However, this method cannot separate multiple instruments of the same type within a recording and is limited to single-channel data. Sarker et al. introduce EnsembleSet, a dual-channel dataset for chamber music generated from MIDI files [23], which retains spatial placement information of instruments. However, subsequent studies merge this dual-channel data into a single channel, failing to fully utilize spatial cues, leading to reduced separation performance in overlapping pitch and similar timbre scenarios [24]. Additionally, both studies only address the separation of two sound sources and do not explore more complex multi-source scenarios.

3. Proposed Method

In Section 2.1, we outline three types of sound source separation methods. The first approach, which involves training a separate model for each instrument, costs many resources. The second approach uses a shared model for all sources but lacks the ability to effectively encode positional information and suffers from performance degradation as the number of outputs increases.

In contrast, the third approach allows dynamic modulation of the network by incorporating positional encoding as a control condition, enabling the network to produce the corresponding instrument audio for a specific target position. Therefore, we adopt the third method for our network design in this study.

3.1. Main Architecture

As shown in Fig. 2, our network is based on a 7-blocked U-Net-like architecture [25] designed for multi-instrument separation using positional information. The input is a complex spectrogram with separate real and imaginary parts, preserving phase information [26] to leverage both magnitude and phase cues for accurate source separation.

The encoder begins with a 1×2 convolution block to adjust the input channel dimension, followed by ReLU for non-linearity. It includes multiple down-sampling layers that reduce the resolution of feature maps, 2D convolutional layers to capture local patterns, and batch normalization layers to stabilize training. The decoder mirrors the encoder with up-sampling layers to restore feature map resolution and skip connections to retain spatial information across scales. The final 1×2 convolution block outputs the target spectrogram, preserving both real and imaginary components. Down-sampling and up-sampling layers handle resolution changes, while smaller convolutional filters capture fine temporal details. For the convolution blocks, we adapt TFC-TDF blocks inspired by [26], as is shown to be able to extract time-frequency features. We set number of internal layers to 5, with kernel size 3×3 densely connected, and the growth rate [27] is set to be 24.

3.2. Timbral and Positional Cue Embedding Module

The timbral and positional cue embedding module encodes the timbral and spatial information of each instrument in the mixture. As shown in the right part of Fig 2, we number the positional candidates and timbral candidates in the string quartets respectively. The position number corresponds to the order in the array ['pos1', 'pos2', 'pos3', 'pos4'] (0, 1, 2 or 3), and the timbre number corresponds to the order in ['Violin', 'Viola', 'Cello'] (0, 1, 2). We pass the values of the positional and timbral information numbers through an embedding generator to generate a 32-dimensional embedding vector, and then multiply the generated weights by 0.5 and add them together. Then the result is used to modulate the weights of the Gated Point-wise Convolutional Modulation(GPoCMs) introduced in section 3.3 at various levels of the decoder. This process conditions the network to focus on the target source. The Condition Generator produces a set of condition weights, $\beta_1, \beta_2, \dots, \beta_C$, which are applied to the decoder's feature maps via GPoCM, dynamically adjusting the network's focus based on the positional cues of the target source.

3.3. Gated Point-wise Convolutional Modulation

Inspired by [28], the Gated Point-wise Convolutional Modulation (GPoCM) is introduced in our work, which extends the idea of Feature-wise Linear Modulation (FiLM) by incorporating a gating mechanism that enhances the flexibility and expressiveness of the modulation process [29]. GPoCM integrates gating mechanisms with inter-channel operations to dynamically control the feature maps at different decoder layers, based on the positional cues embedded by the Condition Generator.

$$PoCM(D_c^i | \omega_c^i, \beta_c^i) = \sum_j \omega_{cj}^i D_j^i + \beta_c^i, \quad (1)$$

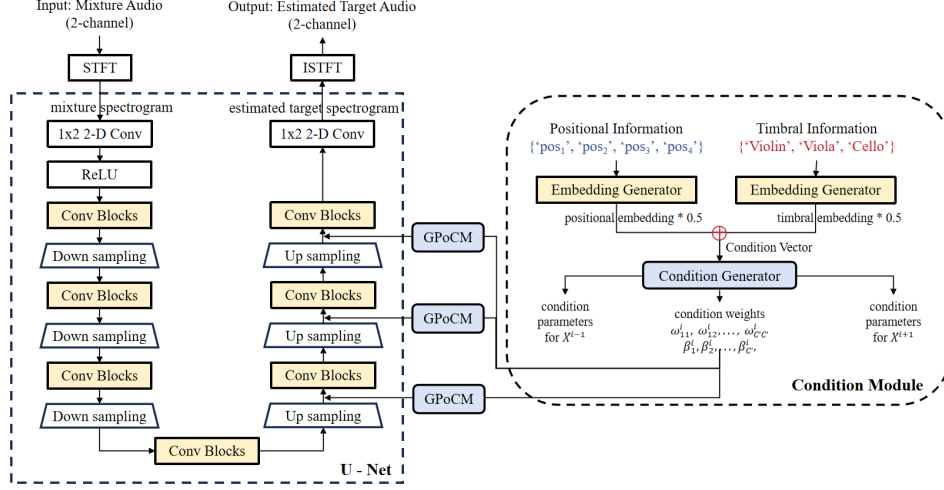


Figure 2: Our proposed method. The left side shows a U-Net structure, and the right side shows the Condition Module. We encode the positional information and timbral information and average the embeddings. Then we generate the condition to control the output of the left U-Net through GPoCM.

where β_c and ω_c represents the position-based weights introduction in 3.2, D_c^i is the output of the i -th decoder. After the channel-wise linear combination, Sigmoid-gated point-wise convolutional modulation is use as

$$GPoCM(D_c^i | \omega_c^i, \beta_c^i) = \sigma(PoCM(D_c^i | \omega_c^i, \beta_c^i)) \odot D_c^i, \quad (2)$$

with σ represents the Sigmoid function and $\odot$ refers to the Hadamard product.

This modulation allows the network to adjust its focus on different spatial regions adaptively, improving its ability to separate target instruments.

4. Experiments

4.1. Dataset

Given that there is currently a lack of string quartet datasets with clean stems and real hall reverberation, we introduce a new dataset: String Quartet with Reverberation Set(SQR Set), with each track is stereo, sampled at 44100 Hz, consisting of the mixture and four sources: *Violin₁*, *Violin₂*, *Viola* and *Cello*. This dataset simulates the spatial situation of two common string quartet seating arrangements in the presence of real hall reverberation. Our dataset has been uploaded to zenodo: <https://zenodo.org/records/14793971>.

Stems Collection: It is difficult to obtain clean stem ground truth for live recorded chamber music string quartets, as bleeding from other instruments cannot be avoided when recording simultaneously. Thus we find string quartet parts from EnsembleSet[23] (synthetic data) and URMP[22] (real recorded data). EnsembleSet is a dataset for chamber music rendered from MIDI files using a highly realistic orchestral sample library by Spitfire Audio called "BBC Symphony Orchestra"(BBCSO), having 18 unique multi-mic recordings and 2 professional mixes. Here we utilize its "Mix_2" parts and take the average value of two channels to generate the single channel data. URMP has mono recordings of isolated instruments in anechoic conditions.

Seating Arrangement: The string quartet consists of two violins, a viola, and a cello. When performing in string quar-

tets, musicians usually sit in a semi-circle so that all of the musicians can see each other. Being able to see each other helps the musicians hear each other so they can better keep time and know where everyone is. However, the order in which they sit has changed over time[30]. Through the 19th century, the musicians were organized in the order of first violin, cello, viola, and second violin, as shown in Fig 1-(a). In contemporary times, the most common seating arrangement for string quartets is shown in Fig 1-(b).

Hall Reverberation Simulation: To simulate real hall reverberation, we take these one-channel audio and create stereo versions in halls using convolution reverb by ambisonic b-format impulse response "s1r2.wav" from the Openair Database¹.

Train, Validation and Test Set: Our training set includes data generated only using 65 songs from the Ensemble-set with seating arrangement type #2, having a total duration of 17002.73 seconds. The validation set is made with the rest 6 songs from the Ensemble-set with type #1 and #2 and duration is $1618.9 * 2 = 3237.8$ seconds. The test set is formulated using URMP with type #1 and #2 and duration is $781.233 * 2 = 1562.466$ seconds. To ensure fairness to each track, we performed RMS normalization on the volume of the four stems in the same song during data synthesis.

4.2. Experimental Setup

For our model, we train it using Adam with learning rate $lr=0.0005$ to minimize the mean squared error between the ground-truth STFT output and the estimated with max epochs of 1000. We don't use any data augmentation methods. For validation and test, we use SDR[31] as the evaluation metric and evaluate the mean absolute error of the target signal and the estimated. We use the median SDR(unit: dB) value over all the test set tracks for each run and report the mean SDR over three runs. For all models in our study, we use FFT window size of 2048 and hop size of 1024 for spectrogram. For the condition module, we tried different combinations of information injection places. We find that we give timbral information to

¹<https://www.openair.hosted.york.ac.uk/>

Table 1: An ablation study: dedicated means U-Nets for single source separation, trained separately. One-conditioned means encoding with only the stem name. 'Ours' is our proposed model. Lines marked with '*' denote results are evaluated with different seating arrangement as training. Otherwise, the same seating arrangement is used. The evaluation method is SDR and unit is dB.

Model	Validation				Test			
	<i>Violin₁</i>	<i>Violin₂</i>	<i>Viola</i>	<i>Cello</i>	<i>Violin₁</i>	<i>Violin₂</i>	<i>Viola</i>	<i>Cello</i>
dedicated	11.68	10.77	10.74	11.32	9.69	3.63	3.14	7.97
dedicated*	8.98	0.18	4.53	0.35	9.65	-0.36	0.57	-0.05
one-conditioned	10.78	10.33	9.47	10.56	2.42	0.73	1.28	5.26
one-conditioned*	8.77	5.79	6.72	4.73	2.80	-0.16	0.16	2.50
ours	11.88	11.83	10.58	13.06	7.94	6.24	7.35	7.63
ours*	11.26	8.26	9.04	8.11	7.85	6.01	7.25	7.57

Table 2: A comparison of SDR performance of our models with other systems. Lines marked with '*' denote results are evaluated with different seating arrangement as training. Otherwise, the same seating arrangement is used. The evaluation method is SDR and unit is dB.

Model	Validation				Test				Model Size
	<i>Violin₁</i>	<i>Violin₂</i>	<i>Viola</i>	<i>Cello</i>	<i>Violin₁</i>	<i>Violin₂</i>	<i>Viola</i>	<i>Cello</i>	
BS-Roformer[6]	17.18	14.07	13.42	15.99	13.35	6.02	5.76	11.81	95M (* 4)
BS-Roformer*	14.29	-0.68	6.24	-0.83	12.89	-0.59	2.92	-0.62	
HTDemucs[3]	14.31	13.09	12.76	14.80	4.39	1.27	2.79	8.93	161M
HTDemucs*	5.44	2.27	0.90	-0.12	5.86	-0.25	-1.96	-0.74	
ours	11.88	11.83	10.58	13.06	7.94	6.24	7.35	7.63	46M
ours*	11.26	8.26	9.04	8.11	7.85	6.01	7.25	7.57	

the encoder, positional information to the decoder and half positional and half timbral information to the middle part has the best performance.

4.3. Results

We perform an ablation study to validate the effectiveness of our proposed method. The group "dedicated" means U-Nets for the single source separation, trained separately. The U-Net structure here is the same as the U-Net setting in our proposed method, but the "Condition Module" and "GPoCM" parts are not added. The "one-conditioned" group has a similar conditional control model to our proposed method, except its condition vector is generated by one condition. Sepcifically, we encode the four tracks by the track name, that is, the four tracks are numbered 0, 1, 2, and 3 respectively according to the order of the names of the four tracks in the array ['*Violin₁*', '*Violin₂*', '*Viola*', '*Cello*']. Results of our ablation study are shown in Table 1. The "validation" group means the results are evaluated on the validation set and the "test" group on the test set. For each model, we evaluated them with the same seating arrangement situation as training(type #2, no mark) and the different(type #1, marked with "*" in the model column). All three methods work well when the data distribution is the same as the training dataset. For all methods, results on the validation set are better than those on the test set. This is due to the difference in data distribution between real recordings and synthesized audio. By observing groups marked with "*", we find the performance of "*Violin₂*" and "*Cello*" in the "dedicated" model trained independently for each instrument dropped significantly, because the positions of these two instruments changed, while the remaining "*Violin₁*" and "*Viola*" drop not that severe. The

"one-conditioned" model performance drops but is better than the dedicated one, indicating that conditional encoding based on the sub-tracks has a certain effect. Our method appears to be more robust.

Also, we compare our proposed method with existing state-of-the-art music source separation models²: BS-Roformer [6] and HTDemucs [3] as mentioned in section 2.1. Each of the contrast method has been trained for 1000 epoch and we confirm that has been fully converged by observing the change in loss. For BS-Roformer, we train one model for each source with batch size of 10 and learning rate of 0.00005. Results are shown in Table 2. Similar to the results in Table 1, SOTA methods achieve good results in the same scenario as training, but they require three to four times the number of parameters of ours and their performances degrade significantly after the data type and acoustic scene change. Our method performs better when the scene changes³.

5. Conclusions

In this work, we propose a method for source separation for string quartets using timbral and positional cues in a conditional modulation network which overcomes the limitations of existing methods. We propose a new dataset of string quartet recordings with realistic reverberation with two different seating arrangements. Experimental results demonstrate the effectiveness of this method in separating individual instruments and highlight the value of spatial cues in music source separation.

²The implementation of these two methods refers to <https://github.com/ZFTurbo/Music-Source-Separation-Training>.

³Audio demos are available at: <https://logicino.github.io/SeparatingSameInstrumentsInStringQuartet/>

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